

MySQL Practical Assignment – Employee Data Analysis

Step 1: Create Database

```
CREATE DATABASE employee_db;  
USE employee_db;
```

Step 2: Create Employees Table

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    department VARCHAR(30) NOT NULL,  
    job_role VARCHAR(50) NOT NULL,  
    salary DECIMAL(10,2),  
    age INT,  
    city VARCHAR(30)  
);
```

Step 3: Insert 20 Employee Records

```
INSERT INTO employees VALUES  
(1,'Achyuth','achyuth@gmail.com','IT','Software  
Developer',55000,24,'Hyderabad'),  
(2,'Priya','priya@gmail.com','HR','HR  
Executive',42000,27,'Vijayawada'),  
(3,'Arjun','arjun@gmail.com','Finance','Accountant',48000,29  
, 'Chennai'),  
(4,'Sneha','sneha@gmail.com','Marketing','Marketing  
Executive',45000, 25,'Bengaluru'),  
(5,'Kiran','kiran@gmail.com','IT','Data  
Analyst',60000,26,'Hyderabad'),  
(6,'Anjali','anjali@gmail.com','HR','Recruiter',40000,24,'Vijayawada'),  
(7,'Vikram','vikram@gmail.com','Finance','Financial  
Analyst',65000,31,'Chennai'),  
(8,'Divya','divya@gmail.com','Marketing','Sales  
Executive',38000,23,'Bengaluru'),  
(9,'Ravi','ravi@gmail.com','IT','Tester',50000,28,'Hyderabad'),  
(10,'Meena','meena@gmail.com','HR','HR  
Manager',70000,35,'Vijayawada'),
```



```
SELECT emp_name,department,salary,city
FROM employees;
```

```
39  
40 • SELECT emp_name, department, salary, city  
41 FROM employees;  
42  
43 • SELECT *
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

emp_name	department	salary	city
Achyuth	IT	55000.00	Hyderabad
Priya	HR	42000.00	Vijayawada
Arjun	Finance	48000.00	Chennai
Sneha	Marketing	45000.00	Bengaluru
Kiran	IT	60000.00	Hyderabad
Anjali	HR	40000.00	Vijayawada
Vikram	Finance	65000.00	Chennai
Divya	Marketing	38000.00	Bengaluru
Ravi	IT	50000.00	Hyderabad
Meena	HR	70000.00	Vijayawada
Suresh	Finance	55000.00	Chennai
Pooja	Marketing	72000.00	Bengaluru
Naveen	IT	65000.00	Hyderabad
Lakshmi	HR	42000.00	Vijayawada
Manoj	Finance	60000.00	Chennai
Swathi	Marketing	40000.00	Bengaluru
Rohit	IT	55000.00	Hyderabad
Neha	HR	40000.00	Vijayawada
Ajay	Finance	48000.00	Chennai
Varun	IT	45000.00	Gannavaram

```
SELECT *
FROM employees
WHERE department = 'IT';
```

```
13 • SELECT *
14 FROM employees
15 WHERE department = 'IT';
```

[illegible]

```
SELECT * FROM employees WHERE salary = 55000;
```

```
47 • SELECT * FROM employees WHERE salary = 55000;
```

[illegible]

```
SELECT * FROM employees WHERE salary > 55000;
```

```
49 • SELECT * FROM employees WHERE salary > 55000;
```

[illegible]

```
SELECT * FROM employees WHERE age < 25;
```

```
51 • SELECT * FROM employees WHERE age < 25;
```

[illegible]

```
SELECT * FROM employees WHERE salary >= 60000;
```

```
53 • SELECT * FROM employees WHERE salary >= 60000;
```

[illegible]

SELECT * FROM employees WHERE salary <= 45000;

56. SELECT * FROM employees WHERE salary <= 45000;

emp_id	emp_name	email	department	job_role	salary	age	city
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada
4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada
8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada
16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram

SELECT * FROM employees WHERE salary <> 55000;

57. SELECT * FROM employees WHERE salary <> 55000;

emp_id	emp_name	email	department	job_role	salary	age	city
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada
3	Arjun	arjun@gmail.com	Finance	Accountant	48000.00	29	Chennai
4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada
7	Vikram	vikram@gmail.com	Finance	Financial Analyst	65000.00	31	Chennai
8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru
9	Ravi	ravi@gmail.com	IT	Tester	50000.00	28	Hyderabad
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada
15	Manoj	manoj@gmail.com	Finance	Financial Analyst	60000.00	30	Chennai
16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada
19	Ajay	ajay@gmail.com	Finance	Accountant	48000.00	28	Chennai
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram

SELECT * FROM employees WHERE salary != 55000;

59. SELECT * FROM employees WHERE salary != 55000;

emp_id	emp_name	email	department	job_role	salary	age	city
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada
3	Arjun	arjun@gmail.com	Finance	Accountant	48000.00	29	Chennai
4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada
7	Vikram	vikram@gmail.com	Finance	Financial Analyst	65000.00	31	Chennai
8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru
9	Ravi	ravi@gmail.com	IT	Tester	50000.00	28	Hyderabad
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada
15	Manoj	manoj@gmail.com	Finance	Financial Analyst	60000.00	30	Chennai
16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada
19	Ajay	ajay@gmail.com	Finance	Accountant	48000.00	28	Chennai
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram

SELECT * FROM employees WHERE department <> 'IT';

51 • **SELECT * FROM employees WHERE department <> 'IT';**

Result Grid								
Filter Rows:								
Edit: Export/Import: Wrap Cell Content:								
emp_id	emp_name	email	department	job_role	salary	age	city	
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada	
3	Arjun	arjun@gmail.com	Finance	Accountant	48000.00	29	Chennai	
4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru	
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada	
7	Vikram	vikram@gmail.com	Finance	Financial Analyst	65000.00	31	Chennai	
8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru	
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada	
11	Suresh	suresh@gmail.com	Finance	Senior Accountant	55000.00	33	Chennai	
12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru	
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada	
15	Manoj	manoj@gmail.com	Finance	Financial Analyst	60000.00	30	Chennai	
16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru	
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada	
19	Ajay	ajay@gmail.com	Finance	Accountant	48000.00	28	Chennai	
* NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

Step 6: Logical Operators

SELECT * FROM employees
WHERE department = 'IT' AND salary > 55000;

63 • **SELECT * FROM employees**
64 **WHERE department = 'IT' AND salary > 55000;**

Result Grid								
Filter Rows:								
Edit: Export/Import: Wrap Cell Content:								
emp_id	emp_name	email	department	job_role	salary	age	city	
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad	
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad	
* NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

SELECT * FROM employees
WHERE department = 'IT' OR department = 'HR';

66 • **SELECT * FROM employees**
67 **WHERE department = 'IT' OR department = 'HR';**

Result Grid								
Filter Rows:								
Edit: Export/Import: Wrap Cell Content:								
emp_id	emp_name	email	department	job_role	salary	age	city	
1	Achyuth	achyuth@gmail.com	IT	Software Developer	55000.00	24	Hyderabad	
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada	
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad	
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada	
9	Ravi	ravi@gmail.com	IT	Tester	50000.00	28	Hyderabad	
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada	
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad	
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada	
17	Rohit	rohit@gmail.com	IT	Software Developer	55000.00	25	Hyderabad	
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada	
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram	
* NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

```
SELECT * FROM employees
WHERE NOT department = 'IT';
```

```
69 • SELECT * FROM employees
70 WHERE NOT department = 'IT';
```

[illegible]

Step 7: Special Operators

```
SELECT * FROM employees
WHERE salary BETWEEN 45000 AND 60000;
```

```
72 • SELECT * FROM employees
73 WHERE salary BETWEEN 45000 AND 60000;
```

7A

[illegible]

```
SELECT * FROM employees
WHERE department IN ('IT', 'HR');
```

```
75 • SELECT * FROM employees
76 WHERE department IN ('IT', 'HR');
77
```

Result Grid								
Filter Rows:								
Edit:								
Export/Import:								
Wrap Cell Content:								
emp_id	emp_name	email	department	job_role	salary	age	city	
1	Achyuth	achyuth@gmail.com	IT	Software Developer	55000.00	24	Hyderabad	
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada	
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad	
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada	
9	Ravi	ravi@gmail.com	IT	Tester	50000.00	28	Hyderabad	
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada	
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad	
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada	
17	Rohit	rohit@gmail.com	IT	Software Developer	55000.00	25	Hyderabad	
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada	
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram	
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

```
SELECT * FROM employees
WHERE city NOT IN ('Hyderabad', 'Chennai');
```

```
78 • SELECT * FROM employees
79 WHERE city NOT IN ('Hyderabad', 'Chennai');
80
```

Result Grid								
Filter Rows:								
Edit:								
Export/Import:								
Wrap Cell Content:								
emp_id	emp_name	email	department	job_role	salary	age	city	
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada	
4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru	
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada	
8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru	
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada	
12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru	
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada	
16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru	
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada	
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram	
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

```
SELECT * FROM employees
WHERE emp_name LIKE 'A%';
```

```
81 • SELECT * FROM employees
82 WHERE emp_name LIKE 'A%';
```

[illegible]

```
SELECT * FROM employees WHERE city IS NULL;
```

```
84 • SELECT * FROM employees
85 WHERE city IS NULL;
86
```

[illegible]

```
SELECT * FROM employees WHERE city IS NOT NULL;
```

```
88 • SELECT * FROM employees
89 WHERE city IS NOT NULL;
```

[illegible]

Step 8: ORDER BY

```
SELECT * FROM employees
ORDER BY salary ASC;
```

```
91 • SELECT * FROM employees
92 ORDER BY salary ASC;
```

emp_id	emp_name	email	department	job_role	salary	age	city
8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada
16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada
4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram
3	Arjun	arjun@gmail.com	Finance	Accountant	48000.00	29	Chennai
19	Ajay	ajay@gmail.com	Finance	Accountant	48000.00	28	Chennai
9	Ravi	ravi@gmail.com	IT	Tester	50000.00	28	Hyderabad
1	Achyuth	achyuth@gmail.com	IT	Software Developer	55000.00	24	Hyderabad
11	Suresh	suresh@gmail.com	Finance	Senior Accountant	55000.00	33	Chennai
17	Rohit	rohit@gmail.com	IT	Software Developer	55000.00	25	Hyderabad
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad
15	Manoj	manoj@gmail.com	Finance	Financial Analyst	60000.00	30	Chennai
7	Vikram	vikram@gmail.com	Finance	Financial Analyst	65000.00	31	Chennai
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru

```
SELECT * FROM employees
ORDER BY salary DESC;
```

```
94 • SELECT * FROM employees
95 ORDER BY salary DESC;
```

Result Grid									Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:	
	emp_id	emp_name	email	department	job_role	salary	age	city								
▶	12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru								
	10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada								
	7	Vikram	vikram@gmail.com	Finance	Financial Analyst	65000.00	31	Chennai								
	13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad								
	5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad								
	15	Manoj	manoj@gmail.com	Finance	Financial Analyst	60000.00	30	Chennai								
	1	Achyuth	achyuth@gmail.com	IT	Software Developer	55000.00	24	Hyderabad								
	11	Suresh	suresh@gmail.com	Finance	Senior Accountant	55000.00	33	Chennai								
	17	Rohit	rohit@gmail.com	IT	Software Developer	55000.00	25	Hyderabad								
	9	Ravi	ravi@gmail.com	IT	Tester	50000.00	28	Hyderabad								
	3	Arjun	arjun@gmail.com	Finance	Accountant	48000.00	29	Chennai								
	19	Ajay	ajay@gmail.com	Finance	Accountant	48000.00	28	Chennai								
	4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru								
	20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram								
	2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada								
	14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada								
	6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada								
	16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru								
	18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada								
	8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru								
◀	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL								

```
SELECT * FROM employees
ORDER BY department ASC, salary DESC;
```

```
97 • SELECT * FROM employees
98 ORDER BY department ASC, salary DESC;
```

emp_id	emp_name	email	department	job_role	salary	age	city
7	Vikram	vikram@gmail.com	Finance	Financial Analyst	65000.00	31	Chennai
15	Manoj	manoj@gmail.com	Finance	Financial Analyst	60000.00	30	Chennai
11	Suresh	suresh@gmail.com	Finance	Senior Accountant	55000.00	33	Chennai
3	Arjun	arjun@gmail.com	Finance	Accountant	48000.00	29	Chennai
19	Ajay	ajay@gmail.com	Finance	Accountant	48000.00	28	Chennai
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
2	Priya	priya@gmail.com	HR	HR Executive	42000.00	27	Vijayawada
14	Lakshmi	lakshmi@gmail.com	HR	HR Executive	42000.00	26	Vijayawada
6	Anjali	anjali@gmail.com	HR	Recruiter	40000.00	24	Vijayawada
18	Neha	neha@gmail.com	HR	Recruiter	40000.00	23	Vijayawada
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad
1	Achyuth	achyuth@gmail.com	IT	Software Developer	55000.00	24	Hyderabad
17	Rohit	rohit@gmail.com	IT	Software Developer	55000.00	25	Hyderabad
9	Ravi	ravi@gmail.com	IT	Tester	50000.00	28	Hyderabad
20	Varun	varun@gmail.com	IT	Support Engineer	45000.00	26	Gannavaram
12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru
4	Sneha	sneha@gmail.com	Marketing	Marketing Executive	45000.00	25	Bengaluru
16	Swathi	swathi@gmail.com	Marketing	Content Writer	40000.00	24	Bengaluru
8	Divya	divya@gmail.com	Marketing	Sales Executive	38000.00	23	Bengaluru
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Step 9: Aggregate Functions

```
SELECT COUNT(*) AS total_employees
FROM employees;
```

```
00 • SELECT COUNT(*) AS total_employees
01 FROM employees;
```

total_employees
20

```
SELECT SUM(salary) AS total_salary
FROM employees;
```

```
03 • SELECT SUM(salary) AS total_salary
04 FROM employees;
```

total_salary
1035000.00

```
SELECT AVG(salary) AS average_salary
FROM employees;
```

```
06 • SELECT AVG(salary) AS average_salary
07 FROM employees;
08
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
average_salary			
51750.000000			

```
SELECT MIN(salary) AS minimum_salary
FROM employees;
```

```
109 • SELECT MIN(salary) AS minimum_salary
110 FROM employees;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
minimum_salary			
38000.00			

```
SELECT MAX(salary) AS maximum_salary
FROM employees;
```

```
12 • SELECT MAX(salary) AS maximum_salary
13 FROM employees;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
maximum_salary			
72000.00			

Step 10: GROUP BY

```
SELECT department, COUNT(*) AS employee_count FROM employees
GROUP BY department;
```

```
115 • SELECT department, COUNT(*) AS employee_count
116 FROM employees
117 GROUP BY department;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
department	employee_count		
IT	6		
HR	5		
Finance	5		
Marketing	4		

SELECT department, AVG(salary) AS average_salary FROM employees
GROUP BY department;

```
L16 • SELECT department, AVG(salary) AS average_salary
L17 FROM employees
L18 GROUP BY department;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
department	average_salary		
IT	55000.000000		
HR	46800.000000		
Finance	55200.000000		
Marketing	48750.000000		

SELECT department, MIN(salary) AS minimum_salary
FROM employees
GROUP BY department;

```
.20 • SELECT department, MIN(salary) AS minimum_salary
.21 FROM employees
.22 GROUP BY department;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
department	minimum_salary		
IT	45000.00		
HR	40000.00		
Finance	48000.00		
Marketing	38000.00		

SELECT department, MAX(salary) AS maximum_salary
FROM employees
GROUP BY department;

```
L24 • SELECT department, MAX(salary) AS maximum_salary
L25 FROM employees
L26 GROUP BY department;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
department	maximum_salary		
IT	65000.00		
HR	70000.00		
Finance	65000.00		
Marketing	72000.00		

SELECT department, SUM(salary) AS total_salary FROM employees
GROUP BY department;

```
.28 • SELECT department, SUM(salary) AS total_salary
.29 FROM employees
.30 GROUP BY department;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
department	total_salary		
IT	330000.00		
HR	234000.00		
Finance	276000.00		
Marketing	195000.00		

Step 11: HAVING

```
SELECT department, COUNT(*) AS employee_count FROM  
employees GROUP BY department HAVING COUNT(*) > 4;
```

```
L32 • SELECT department, COUNT(*) AS employee_count  
L33 FROM employees  
L34 GROUP BY department  
L35 HAVING COUNT(*) > 4;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	department	employee_count			
▶	IT	6			
	HR	5			
	Finance	5			

```
SELECT department, AVG(salary) AS average_salary  
FROM employees GROUP BY department HAVING  
AVG(salary) > 50000;
```

```
L37 • SELECT department, AVG(salary) AS average_salary  
L38 FROM employees  
L39 GROUP BY department  
L40 HAVING AVG(salary) > 50000;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	department	average_salary			
▶	IT	55000.000000			
	Finance	55200.000000			

```
SELECT city, COUNT(*) AS employee_count FROM  
employees GROUP BY city HAVING COUNT(*) > 4;
```

```
L42 • SELECT city, COUNT(*) AS employee_count  
L43 FROM employees  
L44 GROUP BY city  
L45 HAVING COUNT(*) > 4;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	city	employee_count			
▶	Hyderabad	5			
	Vijayawada	5			
	Chennai	5			

```
SELECT department, MAX(salary) AS maximum_salary FROM employees  
GROUP BY department HAVING MAX(salary) > 65000;
```

```
L47 • SELECT department, MAX(salary) AS maximum_salary  
L48 FROM employees  
L49 GROUP BY department  
L50 HAVING MAX(salary) > 65000;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	department	maximum_salary			
▶	HR	70000.00			
	Marketing	72000.00			

Step 12: Challenge Questions

```
SELECT MAX(salary) AS second_highest_salary
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

```
L52 • SELECT MAX(salary) AS second_highest_salary
L53 FROM employees
L54 WHERE salary < (SELECT MAX(salary) FROM employees);
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
second_highest_salary	70000.00			

```
SELECT * FROM employees WHERE salary > (SELECT
AVG(salary) FROM employees);
```

```
.56 • SELECT *
.57 FROM employees
.58 WHERE salary > (SELECT AVG(salary) FROM employees);
```

[illegible]

```
SELECT * FROM employees WHERE salary = (SELECT MAX(salary) FROM employees);
```

```
L60 • SELECT *
L61 FROM employees
L62 WHERE salary = (SELECT MAX(salary) FROM employees);
```

[illegible]

```
SELECT * FROM employees WHERE salary = (SELECT MIN(salary) FROM employees);
```

```
L64 • SELECT *
L65 FROM employees
L66 WHERE salary = (SELECT MIN(salary) FROM employees);
```

[illegible]

SELECT department, AVG(salary) AS average_salary FROM employees
GROUP BY department ORDER BY average_salary DESC LIMIT 1;

```

L65 • SELECT department, AVG(salary) AS average_salary
L66 FROM employees
L67 GROUP BY department
L68 ORDER BY average_salary DESC
L69 LIMIT 1;

```

department	average_salary
Finance	55200.000000

SELECT department, COUNT(*) AS employee_count FROM employees
GROUP BY department ORDER BY employee_count DESC LIMIT 1;

```

L71 • SELECT department, COUNT(*) AS employee_count
L72 FROM employees
L73 GROUP BY department
L74 ORDER BY employee_count DESC
L75 LIMIT 1;

```

department	employee_count
IT	6

SELECT *
FROM employees e
WHERE salary > (
 SELECT AVG(salary)
 FROM employees
 WHERE department = e.department
);

```

L78 • SELECT *
L79 FROM employees e
L80 WHERE salary > (
L81     SELECT AVG(salary)
L82     FROM employees
L83     WHERE department = e.department
L84 );

```

emp_id	emp_name	email	department	job_role	salary	age	city
5	Kiran	kiran@gmail.com	IT	Data Analyst	60000.00	26	Hyderabad
7	Vikram	vikram@gmail.com	Finance	Financial Analyst	65000.00	31	Chennai
10	Meena	meena@gmail.com	HR	HR Manager	70000.00	35	Vijayawada
12	Pooja	pooja@gmail.com	Marketing	Marketing Manager	72000.00	34	Bengaluru
13	Naveen	naveen@gmail.com	IT	Backend Developer	65000.00	27	Hyderabad
15	Manoj	manoj@gmail.com	Finance	Financial Analyst	60000.00	30	Chennai

```

SELECT department, MAX(salary) AS second_highest_salary
FROM employees e
WHERE salary < (
    SELECT MAX(salary)
    FROM employees
    WHERE department = e.department
)
GROUP BY department;

```

```

L86 • SELECT department, MAX(salary) AS second_highest_salary
L87 FROM employees e
L88 WHERE salary < (
L89     SELECT MAX(salary)
L90     FROM employees
L91     WHERE department = e.department
L92 )
L93 GROUP BY department;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
department	second_highest_salary			
IT	60000.00			
HR	42000.00			
Finance	60000.00			
Marketing	45000.00			

```

SELECT city, COUNT(*) AS employee_count
FROM employees
GROUP BY city
ORDER BY employee_count DESC
LIMIT 1;

```

```

L95 • SELECT city, COUNT(*) AS employee_count
L96 FROM employees
L97 GROUP BY city
L98 ORDER BY employee_count DESC
L99 LIMIT 1;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
city	employee_count				
Hyderabad	5				

```

SELECT department, SUM(salary) AS total_salary
FROM employees GROUP BY department HAVING
SUM(salary) > 250000;

```

```

201 • SELECT department, SUM(salary) AS total_salary
202 FROM employees
203 GROUP BY department
204 HAVING SUM(salary) > 250000;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
department	total_salary			
IT	330000.00			
Finance	276000.00			